SIX SHORT ALGEBRAIC NOTES, AFTER GROTHENDIECK'S WORKING NOTES

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Abstract. We state and prove six independent results from Grothendieck's working notes, each a few pages long in the original, and complete or correct them where the pages are incomplete. (1) For $K = \mathbb{Z}[L]/(L-1)^n$, the comparison map from the ring generated by Chern classes to the graded ring of the γ-filtration is surjective, and for $n \geq 3$ it is not injective; its kernel is killed by $(n-1)!$. (2) If $\Omega_{K/k}$ is infinite-dimensional, $K[[t_1, \dots, t_n]]$ with the discrete topology is not formally smooth over K, because no derivation extends $d_{K/k}$. (3) Let A be a finite algebra over a local ring, with an action of a finite group whose order is invertible. Two 1-cocycles with values in A^* that are cohomologous modulo the maximal ideal are cohomologous. There are Sylow and tame-inertia versions. (4) The co-operations on relative integer-valued polynomials are well defined under a flatness condition. (5) For the universal quadratic form of even rank $2n$, $(-1)^n$ det of its polar matrix is congruent modulo 4 to the square of the sum over perfect matchings. (6) A finite, non-empty, connected category in which every connected presheaf has a simply connected category of elements is cofiltered. The case of a finite monoid is the pages' Proposition 1. All the statements proved here are formalised in Lean.

Introduction

The University of Montpellier holds the mathematical papers that Alexandre Grothendieck left to Jean Malgoire and publishes their facsimiles. Alongside long programmes, they contain short computations that settle, or try to settle, a single precise question. This paper collects six of them, from folder 2, folder 22, folder 46, folder 153, folder 77 and folder 158, one per section; the sections can be read independently. A statement is attributed to Grothendieck, with the page, when the page states it. Otherwise it is said to be *after* the page, and we describe the change. The proofs are ours unless we say that we follow the page. Every result is formalised in Lean 4 with Mathlib (Section 7).

1. A Chern ring that is not the graded γ-ring

Folder 2, on the Riemann–Roch theorem, works with integral Chern classes. Page 21 attaches two rings to a λ-ring K with a family of "geometric" elements [5]. The *Chern ring* CK is generated by symbols $c^i(x)$ subject to the Whitney formula, the product rule, and $c^i(E) = 0$ above the rank of a geometric element E. The ring GK is the graded ring of the γ-filtration. A map $\varphi_K \colon CK \to GK$ sends $c^i(x)$ to the class of $\gamma^i(x - \varepsilon(x))$. The page states that φ_K is surjective with torsion kernel, and asks whether it is always injective.

Lemma 1.1 (Grothendieck, folder 2, p. 9)**.** *For $r \geq 1$ and any commutative ring A, in $A[\eta]/(\eta^{r+1})$:*

$$\prod_{i=0}^{r} (1 - i\eta)^{(-1)^i \binom{r}{i}} = 1 + (-1)^{r-1}(r-1)!\,\eta^r.$$

In $K(\mathbb{P}^r) = \mathbb{Z}[\xi]/(\xi^{r+1})$, with $\xi = 1 - L^{-1}$, the class of a point is $\xi^r = \sum_i (-1)^i \binom{r}{i} L^{-i}$, and $c(L^{-i}) = 1 - i\eta$ with η the class of a hyperplane. The lemma therefore computes the total

Date: 7 November 2026.

2020 *Mathematics Subject Classification.* 13-02; 19L10, 13J05, 20J06, 13F20, 11E08, 18A30.

Key words and phrases. Chern classes, lambda-rings, formal smoothness, non-abelian cohomology, integer-valued polynomials, discriminant of quadratic forms, cofiltered categories, Lean.

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Chern class of a point. For $r = 1, 2, 3$ it gives $1 + \eta$, $1 - \eta^2$ and $1 + 2\eta^3$; the factor $(r-1)!$ is the price of integral coefficients.

The page answers its question with $K = \mathbb{Z}[L]/(L-1)^n$, the K-ring of $\mathbb{P}^{n-1}$, in which only L is declared geometric. Then CK is generated by $\zeta = c^1(L)$. Put $\xi = 1 - L^{-1}$; since L is invertible, $\xi^n = 0$ is equivalent to $(L-1)^n = 0$. Following the page, the only relation is $c(\xi^n) = 1$. By Lemma 1.1 this gives $CK = \mathbb{Z}[\zeta]/I$, where I is generated by the $a_m\zeta^m$, $m \geq 1$, and a_m is the coefficient of ζ^m in $\prod_{i=0}^{n}(1 - i\zeta)^{(-1)^i\binom{n}{i}}$. Also $GK = \mathbb{Z}[\zeta]/(\zeta^n)$.

Theorem 1.2 (Grothendieck, folder 2, p. 21, for $n \geq 3$; the case $n \leq 2$ is ours)**.** *For $n \geq 3$, ζ^n is non-zero in CK and $(n-1)!\,\zeta^n = 0$. The ring GK is torsion-free, and φ_K is surjective but not injective; its kernel is killed by $(n-1)!$. For $n \leq 2$, φ_K is bijective.*

Proof. By Lemma 1.1, $a_m = 0$ for $1 \leq m < n$ and $a_n = \pm(n-1)!$. Hence $I \subseteq (\zeta^n)$, the map φ_K, $\zeta \mapsto \zeta$, is well defined and surjective, and $(n-1)!\,\zeta^n = 0$ in CK. The kernel of φ_K is $(\zeta^n)/I$, so it is killed by $(n-1)!$. If $k \neq 0$ and ζ^n divides kf in $\mathbb{Z}[\zeta]$, then the coefficients of f of degree $< n$ vanish, so GK is torsion-free. To see that $\zeta^n \neq 0$ in CK, send ζ to h in $T = (\mathbb{Z}/(n-1)!)[h]/(h^{n+1})$. The relations hold there: $a_m h^m = 0$ for $m > n$ because $h^{n+1} = 0$, and $a_n h^n = 0$ because $(n-1)! = 0$ in T. But $h^n \neq 0$ in T as soon as $(n-1)! > 1$, that is, $n \geq 3$. For $n \leq 2$, $a_n = \pm 1$, so $I = (\zeta^n)$. □

The page asserts that CK has torsion for $n \geq 3$ but does not show that $\zeta^n \neq 0$; the map to T supplies this. The presentation of CK by ζ and the single relation $c(\xi^n) = 1$ is the page's reduction, and we take it as given.

2. Power series and the discrete topology

Folder 22 gathers complements to EGA 0_{IV} (1963 to about 1967). Its typescript, which he numbered 24 to 32, proves the noetherian case of Chase's theorem [3]: if every finitely generated A-module is finitely presented, every product of flat A-modules is flat, because $M \otimes_A \prod_i P_i \to \prod_i (M \otimes_A P_i)$ is an isomorphism for M finitely presented. Handwritten pages 22 to 24 then ask whether $A = k[[t]]$, with the discrete topology, is formally smooth over k. Page 22 expects that it never is when $k = \mathbb{Q}(x_i : i \in I)$ with I infinite, and page 23 states the general case.

Formal smoothness for the discrete topology is the usual lifting property through square-zero ideals; equivalently, $\Omega_{A/K}$ is projective and $H_1(L_{A/K}) = 0$. We use one consequence. If A is formally smooth over K and $k \to K$ is a ring map, the Jacobi–Zariski sequence

$$H_1(L_{A/K}) \to A \otimes_K \Omega_{K/k} \xrightarrow{\iota} \Omega_{A/k} \to \Omega_{A/K} \to 0$$

is exact with $H_1(L_{A/K}) = 0$ and $\Omega_{A/K}$ projective, so ι has an A-linear retraction r. Then $D = r \circ d_{A/k}$ is a k-derivation $A \to A \otimes_K \Omega_{K/k}$ with $D(c) = 1 \otimes dc$ for $c \in K$.

Lemma 2.1. *Let K be a non-zero commutative ring, M a free K-module with basis $(b_i)_{i \in I}$, $(i_n)_{n \geq 0}$ an injective sequence in I and (a_n) a sequence in K. There is no additive map $D\colon K[[t]] \to K[[t]] \otimes_K M$ satisfying the Leibniz rule with $D(a_n) = 1 \otimes b_{i_n}$ for all n.*

Proof. For $j \in I$, let $p_j\colon K[[t]] \otimes_K M \to K[[t]]$ be the $K[[t]]$-linear map $f \otimes m \mapsto b_j^*(m)\, f$. An element of the tensor product is a finite sum of pure tensors, so only finitely many of its coordinates p_j are non-zero. Put $f = \sum_n a_n t^n$, and choose n with $p_{i_n}(D(f)) = 0$ and $p_{i_n}(D(t)) = 0$. Since p_{i_n} is $K[[t]]$-linear, $E = p_{i_n} \circ D$ is a derivation $K[[t]] \to K[[t]]$. It satisfies $E(t) = 0$, hence $E(t^m g) = t^m E(g)$ for all m and g, and $E(a_m) = \delta_{mn}$ because $m \mapsto i_m$ is injective. Writing $f = \sum_{m \leq n} a_m t^m + t^{n+1} g$ gives $E(f) = t^n + t^{n+1} E(g)$, whose coefficient of t^n is $1 \neq 0$. This contradicts $E(f) = 0$. □

Theorem 2.2 (Grothendieck, folder 22, p. 23)**.** *Let k be a commutative ring and K a field that is a k-algebra, with $\Omega_{K/k}$ infinite-dimensional over K. Let $A = K[[t_1, \ldots, t_n]]$ with $n \geq 1$. No k-derivation $D\colon A \to A \otimes_K \Omega_{K/k}$ satisfies $D(c) = 1 \otimes dc$ for all $c \in K$. Hence A, with the discrete topology, is not formally smooth over K.*

Proof. The dc, $c \in K$, span $\Omega_{K/k}$; extract a basis from them, which is infinite by hypothesis, and choose $a_0, a_1, \ldots \in K$ such that $da_0, da_1, \ldots$ are distinct vectors of this basis. For $n = 1$, Lemma 2.1 excludes D. For general n, let $\iota\colon K[[t]] \to A$ send t to t_1, and let $\pi\colon A \to K[[t]]$ send t_1 to t and the other t_i to 0, so that $\pi \circ \iota = \mathrm{id}$. Put $\varphi = \pi \otimes \mathrm{id}$, so that $\varphi(a\tau) = \pi(a)\varphi(\tau)$. If D existed, $\varphi \circ D \circ \iota$ would be a derivation of $K[[t]]$ that induces d on K, which the case $n = 1$ excludes. The last assertion follows from the retraction above. □

The proof is ours. Pages 22 and 23 first reduce to $D(t) = 0$ by shrinking the field of definition; choosing n large enough makes this unnecessary. Page 24 starts the series $\sum_n a_n t^n$ and stops. The condition it keeps on the a_n, $da_n \neq 0$, is not enough (take all a_n equal), and the lemma uses instead the condition the page struck out: a non-zero coordinate of index i_n. Characteristic 0 plays no part. Pages 26 and 28 show that differentials need not inject in characteristic p.

Proposition 2.3 (Grothendieck, folder 22, pp. 26 and 28)**.** *Let p be a prime, $K = \mathbb{F}_p(x, b_0, \ldots, b_{p-1})$, $a_i = b_i^p$, $y = \sum_{i<p} a_i x^i$, $k = \mathbb{F}_p(x, y) \subset K$ and $k' = k \cap K^p$. Then $K \otimes_k \Omega_{k/k'} \to \Omega_{K/k'}$ is not injective: with $y' = \sum_i i a_i x^{i-1}$, the element $1 \otimes dy - y' \otimes dx$ is a non-zero element of the kernel.*

Proof. Every derivation of K kills p-th powers, so $dy = y'\,dx$ in $\Omega_{K/k'}$ and the element lies in the kernel. To see that it is non-zero, it suffices to show that $\partial/\partial y$, the partial derivative of $k = \mathbb{F}_p(x, y)$, kills k'; it then gives a K-linear form on $K \otimes_k \Omega_{k/k'}$ equal to 1 on the element. For $c \in k'$, the partial derivative ∂_x of K kills c, so the chain rule gives $0 = \partial c/\partial x + y'\,\partial c/\partial y$. Since x, y, y' are algebraically independent over $\mathbb{F}_p$, $y' \notin k$, and therefore $\partial c/\partial y = 0$. □

Since $k' \subseteq K^p$, $\Omega_{K/k'} = \Omega_{K/\mathbb{F}_p}$, which is the target the page writes. Page 28 proves by contradiction that (x, y) is p-free over k'; the proof above obtains this from $\partial/\partial y$.

3. Lifting cocycles through the reduction modulo $\mathfrak{m}$

Folder 46 ends its "petits fourbis cohomologiques" (pages 71 to 92) with a semi-local Hilbert 90. Let $\mathcal{O}$ be a commutative local ring with maximal ideal $\mathfrak{m}$ and residue field of characteristic p, Γ a finite group, and A an $\mathcal{O}$-algebra, not necessarily commutative, on which Γ acts by ring automorphisms; the margin of page 90 cites $A = \mathrm{M}_n(\mathcal{O})$. A 1-cocycle is a map $c\colon \Gamma \to A^*$ with $c(gh) = c(g)\,g(c(h))$. Two cocycles c, c' are cohomologous if $c(g) = u^{-1}c'(g)\,g(u)$ for some $u \in A^*$. For A non-commutative, $H^1(\Gamma, A^*)$ is only a pointed set, and a trivial kernel does not imply injectivity.

Theorem 3.1 (after folder 46, pp. 90–92)**.** *Suppose $p \nmid |\Gamma|$, and that $\mathfrak{m}^N A = 0$ for some N or A is finite over $\mathcal{O}$. Let $\pi\colon A \to B$ be a surjective Γ-equivariant ring map with kernel in $\mathfrak{m}A$, for instance $B = A/\mathfrak{m}A$. Then $H^1(\Gamma, A^*) \to H^1(\Gamma, B^*)$ is injective: two cocycles with cohomologous reductions are cohomologous.*

Proof. For $x \in \mathfrak{m}A$, $1 + x$ is a unit: by nilpotence when $\mathfrak{m}^N A = 0$, and by Nakayama's lemma, applied to left and right multiplication by $1 + x$, when A is finite. Hence an element of A whose image in B is a unit is itself a unit: if $ww' = 1 + x$ and $w'w = 1 + x'$ with $x, x' \in \ker \pi$, then w has a right and a left inverse. Let $v \in B^*$ with $\pi(c(g)) = v^{-1}\pi(c'(g))\,g(v)$ for all g. Lift v to $w \in A$ and put

$$u = \sum_{\sigma \in \Gamma} c'(\sigma)\,\sigma(w)\,c(\sigma)^{-1}.$$

The cocycle relation gives $g(c(\sigma))^{-1} = c(g\sigma)^{-1}c(g)$, hence

$$c'(g)\,g(u) = \sum_{\sigma} c'(g\sigma)\,(g\sigma)(w)\,c(g\sigma)^{-1}c(g) = u\,c(g).$$

In B, every term of the sum is $\pi(c'(\sigma))\,\sigma(v)\,\pi(c(\sigma))^{-1} = v$, so $\pi(u) = |\Gamma|\,v$ (Figure 1). This is a unit, because $p \nmid |\Gamma|$ makes $|\Gamma|$ invertible in $\mathcal{O}$. Hence $u \in A^*$ and $c(g) = u^{-1}c'(g)\,g(u)$. □

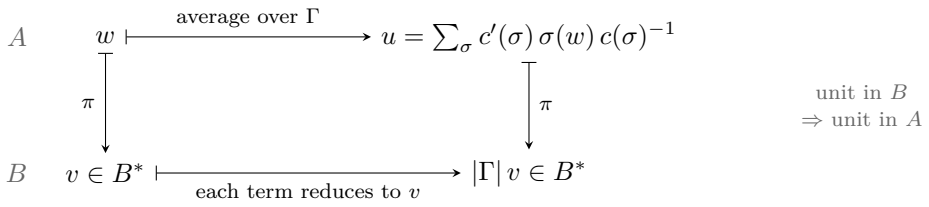


FIGURE 1. The averaging in the proof of Theorem 3.1. A lift w of the coboundary v is averaged against the two cocycles. The average u intertwines c and c', and its reduction is $|\Gamma|\, v$. Since $\ker \pi \subseteq \mathfrak{m}A$, u is a unit.

With $c' = 1$, this is the argument of Hilbert's Theorem 90. The page argues differently. It corrects the cocycle one layer $\mathfrak{m}^n A/\mathfrak{m}^{n+1}A$ at a time, where the cocycle relation becomes additive, and uses Artin–Rees when $\mathfrak{m}$ is not nilpotent on A. It states the case $p \nmid |\Gamma|$ "in particular", after a Sylow version, which we prove in the following form. For commutative A, the trivial-kernel part of Theorem 3.1 is contained in a lemma of Lorenz on $H^1(G, \mathrm{GL}_n(S)) \to H^1(G, \mathrm{GL}_n(S/\operatorname{rad} S))$ ([7], §3.6 in the numbering of the preprint arXiv:math/9808032).

Theorem 3.2 (after folder 46, p. 90 and its margin)**.** *Let $\Lambda \leq \Gamma$ be a subgroup of index prime to p, for instance a Sylow p-subgroup. Let A be finite over $\mathcal{O}$ and $\pi\colon A \to B$ as above, with $A^\Lambda \to B$ surjective; the margin's hypothesis that $A^\Gamma \to A/\mathfrak{m}A$ is surjective implies this. A class of $H^1(\Gamma, A^*)$ that is trivial in $H^1(\Gamma, B^*)$ and in $H^1(\Lambda, A^*)$ is trivial. If A is commutative, $H^1(\Gamma, A^*) \to H^1(\Gamma, B^*) \times H^1(\Lambda, A^*)$ is injective.*

Sketch. Average over Γ/Λ instead of Γ. Let $c(\lambda) = u^{-1}\lambda(u)$ on Λ and $\pi(c(g)) = \bar{v}^{-1}g(\bar{v})$. Choose $w \in A^\Lambda$ lifting $\pi(u)\bar{v}^{-1}$ and put $z = u^{-1}w$. Then $c(\lambda)\lambda(z) = z$ for $\lambda \in \Lambda$, so $c(g)\, g(z)$ depends only on $g\Lambda$, and $b = \sum_{g\Lambda} c(g)\, g(z)$ satisfies $c(\gamma)\gamma(b) = b$. It reduces to $[\Gamma : \Lambda]\, \bar{v}^{-1}$, a unit, so $c(g) = b\, g(b)^{-1}$. For commutative A, H^1 is a group and a trivial kernel suffices. □

For non-commutative A the same averaging compares two cocycles c, c' only if w can be chosen invariant under the action of Λ twisted by c'. This holds when c' is a coboundary on Λ, but we do not know whether the margin's untwisted hypothesis suffices in general.

Pages 84 to 89 treat a base on which Γ acts. The inertia group Γ_i is the subgroup acting trivially on the residue field k, and the action on A is semi-linear: $\gamma(\lambda a) = \gamma(\lambda)\gamma(a)$ for $\lambda \in \mathcal{O}$.

Theorem 3.3 (after folder 46, pp. 84–89)**.** *Let $\mathcal{O}$ be a commutative local ring with infinite residue field k, Γ a finite group acting on $\mathcal{O}$ whose inertia group Γ_i has order invertible in k, and A a finite $\mathcal{O}$-algebra, not necessarily commutative, with a semi-linear action of Γ. Then $H^1(\Gamma, A^*) \to H^1(\Gamma_i, A^*)$ is injective. In particular, if Γ acts faithfully on k, then $H^1(\Gamma, A^*) = 1$; for $\mathfrak{m} = 0$ this is Hilbert's Theorem 90 for any finite algebra over an infinite field.*

Sketch. We may assume $c = c'$ on Γ_i. Work in $B = A/\mathfrak{m}A$ with the twisted semi-linear action $\gamma \star x = c'(\gamma)\gamma(x)c(\gamma)^{-1}$, and let X be its set of fixed points. A Speiser-type lemma, using Dedekind's independence of characters and $|\Gamma_i| \neq 0$ in k, shows that the Γ_i-fixed vector 1 lies in the k-span of X: $1 = \sum_j t_j e_j$ with $e_j \in X$. The determinant of left multiplication by $\sum_j x_j e_j$ is a polynomial in the x_j that takes the value 1 at (t_j). Since k^Γ is infinite, it does not vanish identically on $(k^\Gamma)^n$, and a k^Γ-combination of the e_j is a unit lying in X. Hence $\pi(c)$ and $\pi(c')$ are cohomologous. To lift this to A, Dedekind's lemma gives $\lambda \in \mathcal{O}$ with $\sum_{\sigma\in\Gamma} \sigma(\lambda) = 1$, and the average $\sum_{\sigma\in\Gamma} c'(\sigma)\, \sigma(\lambda w)\, c(\sigma)^{-1}$ takes the place of the average in Theorem 3.1. □

The page goes through $\mathcal{O}^{\Gamma_i}$ and inflation–restriction, takes for granted that Γ/Γ_i acts faithfully on the residue field of $\mathcal{O}^{\Gamma_i}$, and then works layer by layer as before. When the inertia is tame and k is infinite, the proof above needs none of these steps. The same averaging, with a Chinese-remainder element in place of λ, also proves the semi-local version (a base on whose

maximal ideals Γ acts transitively) when the decomposition group has an element of trace one modulo $\mathfrak{m}$. The wild case and a finite residue field, where the page uses Lang's theorem, are not proved here.

4. Co-operations on relative integer-valued polynomials

Folder 153, *Analyseurs*, looks for the abstract structure of a ring of operations with a composition, now called a plethory [8, 1], and reaches it through integer-valued polynomials [2]. Page 3 fixes an inclusion of commutative rings $k_0 \subseteq K_0$ and asks that $F(X+Y)$ and $F(XY)$, for $F \in \Omega$, be "well-determined" elements of $\Omega \otimes_{k_0} \Omega$. The inclusion is not needed. Let K_0 be a commutative k_0-algebra with structure map ι, and put

$$\Omega = \{F \in K_0[T] \mid F(\iota(\lambda)) \in \iota(k_0) \text{ for all } \lambda \in k_0\},$$

a k_0-subalgebra of $K_0[T]$. Let $\beta\colon K_0 \otimes_{k_0} \Omega \to K_0[T]$, $a \otimes F \mapsto aF$, and $\Phi\colon \Omega \otimes_{k_0} \Omega \to K_0[X,Y]$, $G \otimes H \mapsto G(X)H(Y)$.

Theorem 4.1 (after folder 153, pp. 3 and 9)**.** *Let K_0 be any commutative k_0-algebra.*

1. *β is surjective.*
2. *If β is injective and Ω is free over k_0, then Φ is injective, and its image is the set of $P \in K_0[X,Y]$ with $P(\iota(x), \iota(y)) \in \iota(k_0)$ for all $x, y \in k_0$. For $F \in \Omega$, $F(X+Y)$ and $F(XY)$ are the images of unique elements of $\Omega \otimes_{k_0} \Omega$, so Ω carries a co-addition and a co-multiplication.*
3. *β is injective when K_0 is a localisation of k_0 and Ω is free, and when K_0 is flat over k_0 with $K_0 \otimes_{k_0} K_0 = K_0$, that is, with injective multiplication $K_0 \otimes_{k_0} K_0 \to K_0$.*

Sketch. (1) The monomials T^n lie in Ω and span $K_0[T]$ over K_0. (2) Let (e_j) be a basis of Ω. The $1 \otimes e_j$ form a basis of $K_0 \otimes \Omega$, so the surjectivity and injectivity of β say that the e_j span $K_0[T]$ and are K_0-linearly independent. Write $t \in \Omega \otimes \Omega$ as $\sum_j G_j \otimes e_j$. Independence gives the injectivity of Φ, coefficient by coefficient in X. For the image, spanning writes $P = \sum_j G_j(X) e_j(Y)$ with $G_j \in K_0[T]$. For each $x \in k_0$, $P(\iota(x), T) \in \Omega$, and independence gives $G_j(\iota(x)) \in \iota(k_0)$, so $G_j \in \Omega$. The polynomials $F(X+Y)$ and $F(XY)$ take values in $\iota(k_0)$. (3) For a localisation $S^{-1}k_0$, clear denominators in a relation $\sum_j a_j e_j = 0$. In the flat case, β is $\mathrm{id} \otimes (\Omega \hookrightarrow K_0[T])$, injective by flatness, followed by $K_0 \otimes_{k_0} K_0[T] \to K_0[T]$, which is the multiplication of $K_0 \otimes_{k_0} K_0$ taken coefficient by coefficient. □

For $k_0 = \mathbb{Z}$ and $K_0 = \mathbb{Q}$, $\Omega = \mathrm{Int}(\mathbb{Z})$ has the binomial basis $\binom{T}{n}$ of page 2, and Vandermonde's identity gives the co-addition $\binom{T}{m} \mapsto \sum_{i+j=m} \binom{T}{i} \otimes \binom{T}{j}$. This is the classical biring structure of $\mathrm{Int}(\mathbb{Z})$. Elliott [4] studies when $\mathrm{Int}(D)$ is a plethory for a domain D, under the condition that $\mathrm{Int}(D) \otimes_D \mathrm{Int}(D) \to \mathrm{Int}(D^2)$ be an isomorphism; part (2) is the relative analogue of that condition. The page assumes only $K_0 \otimes_{k_0} K_0 = K_0$. We do not know whether this, together with the freeness of Ω, suffices without flatness.

5. The universal discriminant modulo 4

Folder 77, part IV, takes the universal quadratic form $q = \sum_i Y_i x_i^2 + \sum_{i<j} Y_{ij} x_i x_j$ in m variables over $\mathbb{Z}[Y]$, and the determinant of its polar matrix P_m, which has $2Y_i$ on the diagonal and Y_{ij} off it. Pages 65 to 67 expand $\det P_m$ as a sum over graphs. Page 68 shows that $\det P_m \equiv 0 \bmod 2$ for m odd. For $m = 2n$ it puts

$$B_{2n} = \sum_M \prod_{\{i,j\} \in M} Y_{ij},$$

the sum over the perfect matchings M of $\{1, \dots, 2n\}$ (the hafnian of the matrix (Y_{ij})). It then looks for $\varepsilon = \pm 1$ with $B_{2n}^2 - \varepsilon \det P_{2n} \equiv 0 \bmod 4$, and expects $\varepsilon = (-1)^n$.

Theorem 5.1 (Grothendieck, folder 77, pp. 68–71)**.** *For every $n \geq 0$, 4 divides $(-1)^n \det P_{2n} - B_{2n}^2$ in $\mathbb{Z}[Y]$. Hence there is $C_{2n} \in \mathbb{Z}[Y]$ with $(-1)^n \det P = B^2 - 4C$ for every quadratic form of rank $2n$ over every commutative ring. The other sign fails: for $n = 1$ and $n = 2$, $B^2 + (-1)^n \det P$ is not divisible by 4 in general.*

Proof. Write $\det P = \sum_\sigma \operatorname{sgn}(\sigma) \prod_i P_{i\sigma(i)}$ and read each permutation as a graph on $\{1, \dots, 2n\}$: fixed points, transpositions (double edges) and cycles of length at least 3 (Figure 2). Let σ have f fixed points and c cycles of length ≥ 3. Reversing any subset of these cycles gives 2^c permutations with the same cycle type, hence the same sign, and, since P is symmetric, the same product. Each fixed point contributes a factor $2Y_i$. The orbit of σ therefore contributes $\pm 2^{f+c}$ times a monomial, which is 0 modulo 4 unless $f + c \leq 1$. Since $2n$ is even, $f = 1$ and $c = 0$ is impossible. Two kinds of orbits remain.

(a) $f = c = 0$: the fixed-point-free involutions, that is, the perfect matchings M, with sign $(-1)^n$ and product $\prod_{\{i,j\} \in M} Y_{ij}^2$.

(b) $f = 0$, $c = 1$: transpositions together with one cycle, necessarily of even length $2\ell \geq 4$. The orbit $\{\sigma, \sigma^{-1}\}$ contributes ± 2 times the product of Y_e over the edges e of the graph, a double edge counting twice; and $2 \equiv -2 \bmod 4$.

So $(-1)^n \det P \equiv \sum_M \prod_M Y_{ij}^2 + 2S \bmod 4$, where S is the sum of the monomials of the graphs of type (b). On the other side,

$$B^2 = \sum_M \prod_M Y_{ij}^2 + 2 \sum_{\{M, M'\}} \prod_M Y_{ij} \prod_{M'} Y_{ij},$$

the second sum running over unordered pairs of distinct matchings. The union $M \cup M'$ consists of double edges (common edges) and $c \geq 1$ alternating cycles of even length ≥ 4, and its monomial is the product of the Y_{ij} in the pair. Conversely, such a graph arises from exactly 2^c ordered pairs, because each alternating cycle splits into two matchings in two ways. The graphs with $c \geq 2$ thus contribute $2 \cdot 2^{c-1}$ times their monomial, which is 0 modulo 4. The graphs with $c = 1$ arise from exactly one unordered pair, and they are precisely the graphs of type (b), with the same monomial. Hence $B^2 \equiv \sum_M \prod_M Y_{ij}^2 + 2S \equiv (-1)^n \det P \bmod 4$. The polynomial C_{2n} specialises to every ring. For the other sign: with $n = 1$, $a_1 = a_2 = 0$ and $b_{12} = 1$, $\det P = -1$ and $B^2 - \det P = 2$; with $n = 2$, $b_{12} = b_{34} = 1$ and all other coefficients 0, $\det P = 1$ and $B^2 + \det P = 2$. □

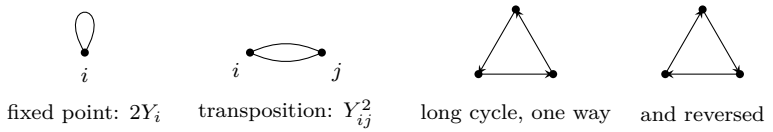


FIGURE 2. The three kinds of cycles in the expansion of $\det P$. A fixed point carries a factor 2. A cycle of length ≥ 3 and its reverse give the same term, so together they carry a factor 2. Modulo 4, only graphs with at most one such factor survive.

Figure 3 shows the case $n = 2$. In rank 2, $-\det P = b_{12}^2 - 4a_1a_2$ is the usual discriminant. In rank 4, C_4 can be written out, and its coefficients agree with those that page 87 checks. The alarm in brown ink on the pages, corrected two lines below, concerned this comparison.

The proof above is our write-up of the pages' method, the graph expansion followed by pairs of matchings. The formal proof goes another way. With U the upper-triangular part of P_{2n} (diagonal Y_i) and $K = U - {}^tU$, a first-order expansion of $\det(P - t\,{}^tU)$ at $t = 2$ gives $\det K \equiv (1 - 2n) \det P \equiv (-1)^n \det P \bmod 4$. Since K is antisymmetric of even size and $\mathbb{Z}[Y]$ is integrally closed, $\det K = r^2$ with $r \in \mathbb{Z}[Y]$ (r is the Pfaffian of K up to sign). Modulo 2, K is symmetric with zero diagonal, so $\det K \equiv B^2$, and therefore $r \equiv B \bmod 2$ and $r^2 \equiv B^2 \bmod 4$. For discriminants of quadratic forms over rings, see Knus [6].

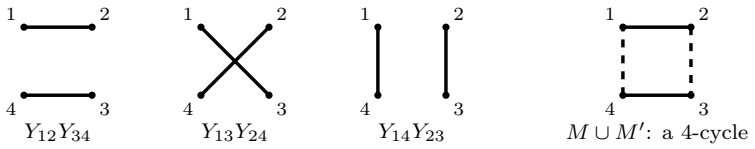


FIGURE 3. Rank 4: the three perfect matchings of $\{1,2,3,4\}$, whose monomials add up to B_4, and the union of two of them, an alternating 4-cycle. Its two orientations are two 4-cycle terms of $\det P$, which together give $-2\,Y_{12}Y_{23}Y_{34}Y_{14}$. Modulo 4 this equals the cross term $2\,Y_{12}Y_{34} \cdot Y_{14}Y_{23}$ of B_4^2.

6. Finite categories with simply connected elements

Folder 158, *Autour des Dérivateurs*, dated about 1990–1991, studies one question. Is a small category X cofiltered as soon as condition (H) holds, namely that every connected presheaf on X has a simply connected category of elements? A presheaf is connected when its category of elements is. That category is simply connected when every functor from it to a group, that is, every cocycle, is isomorphic to the trivial one. A category X is *cofiltered* if it is non-empty, any two objects are targets of arrows from a common object, and for any parallel arrows $f, g\colon x \rightrightarrows y$ there is $h\colon w \to x$ with $fh = gh$. Pages 44 to 53 treat finite categories, starting with monoids, which are categories with one object. A monoid M is *pseudo-cofiltered* if for all u, v there are u', v' with $uu' = vv'$. It is *cofiltered* if moreover for all u, v there is w with $uw = vw$.

Lemma 6.1. *In a finite pseudo-cofiltered monoid M, some p lies in every principal right ideal uM.*

Proof. If $uu' = vv' = w$, then $wM \subseteq uM \cap vM$. By induction, a finite intersection of principal right ideals contains a principal right ideal, and M is finite. □

Theorem 6.2 (Grothendieck, folder 158, Proposition 1, pp. 44–46)**.** *Let M be a finite pseudo-cofiltered monoid whose enveloping group is trivial, that is, every monoid map from M to a group is trivial. Then there is $p \in M$ with $up = p$ for all $u \in M$, and M is cofiltered.*

Proof. Take p as in Lemma 6.1 and put $E_0 = pM$, the orbit of p under the action of M on itself by left multiplication $x \mapsto ux$. Let $u \in M$. The lemma applied to up gives v with $p = upv$, so $py = u(p(vy))$ for all y, and $E_0 \subseteq uE_0$. Since $E_0 \subseteq M$ is finite and uE_0 has at most as many elements as E_0, $uE_0 = E_0$ and $x \mapsto ux$ is a permutation of E_0. This defines a monoid map from M to the symmetric group of E_0. By hypothesis it is trivial, so $ux = x$ for all $x \in E_0$, in particular $up = p$ since $p = p \cdot 1 \in E_0$. Finally $uw = vw$ with $w = p$. □

The page follows the same route: pseudo-cofiltrance, a representation of M by maps of a set, and the step "since E_0 is finite, $u(E_0) = E_0$". The choice $E_0 = pM$ and Lemma 6.1 are ours. The finiteness step holds because E_0 lies in M; before the set E the page struck out the word "finite". The element p need not be unique, and E_0 need not be a point: in the monoid $\{1, a, b\}$ with $xa = a$ and $xb = b$ for all x, both a and b qualify, and $aM = \{a, b\}$. A paragraph struck out under the statement claimed uniqueness, and this example refutes it.

For a monoid with a minimal element (an element lying in every uM), the corollary of page 39 states that (H) implies cofiltered. Conversely, and this statement is ours, for every cofiltered monoid the category of elements of a connected M-set is simply connected.

Theorem 6.3 (after folder 158, p. 52)**.** *In any small category, (H) implies that two elements of a connected presheaf come from a common element. In a non-empty connected category satisfying (H), any two arrows $u\colon x \to c$, $v\colon y \to c$ complete to a square $uu' = vv'$ (Figure 4). A finite, non-empty, connected category satisfying (H) is cofiltered. Connectedness is needed: the discrete category with two objects satisfies (H) and is not cofiltered.*

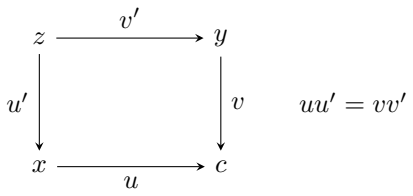


FIGURE 4. The square of Theorem 6.3: the elements u and v of the representable presheaf $\mathrm{Hom}(-, c)$ have a common ancestor $uu' = vv'$.

Sketch. If two elements ξ, η of a connected presheaf F had no common ancestor, the subpresheaves T_1, T_2 they generate would be disjoint. Glue two copies of F along $T_1 \sqcup T_2$. The result is connected, and counting 1 for each arrow from an element of T_1 into the second copy outside $T_1 \sqcup T_2$ defines a $\mathbb{Z}$-valued cocycle. A zigzag from ξ to η shows that this cocycle is not a coboundary. Applied to the terminal presheaf, this gives common sources for pairs of objects; applied to $\mathrm{Hom}(-, c)$, it gives the squares. For a finite X, these give an object e with an arrow to every object. The monoid $\mathrm{End}(e)$ is finite and pseudo-cofiltered, and Lemma 6.1 and a power give an idempotent p lying in every $u\,\mathrm{End}(e)$. Let $L(z) = \{f\colon e \to z \mid fp = f\}$. For $k\colon z \to z'$, composition maps $L(z)$ onto $L(z')$. Call z least if every z' with an arrow $z' \to z$ also receives an arrow from z. Between least objects these surjections of finite sets go both ways, so they are bijections. The least objects form the category of elements of a connected subpresheaf of the terminal presheaf, so (H) makes this local system constant, and that forces $fcp = gcp$ for any $f, g\colon x \rightrightarrows y$ and $c\colon e \to x$. In the discrete category with two objects every arrow is an identity, so every cocycle is trivial. □

The proof of the finite case follows pages 50 to 52; page 53 asserts the step from (H) to squares without argument. As written on page 52, the corollary is stated without connectedness, and the discrete example shows that this hypothesis is needed. Whether (H) implies cofiltered for every small category, the question of the folder, is not settled here.

7. VERIFICATION

The statements proved above are formalised in Lean 4 with Mathlib, using only Lean's standard axioms, in the directory `lean/Grothendieck` of the repository Commutator-IO/grothendieck-project. The main theorems are as follows.

- `Folder2.lean`: `chernPoint` (Lemma 1.1); `GK_torsionFree`, `exists_chern`, `hT_pow_ne_zero`, `ζ_pow_ne_zero`, `factorial_mul_ζ_pow`, `φ_not_injective` and `φ_bijective_of_le_two` (Theorem 1.2).
- `Folder22.lean`: `corollaire3_4`, `corollaire3_4_coherent` (Chase's theorem as in the typescript); `no_derivation` (Lemma 2.1); `retraction_of_formallySmooth`; `no_derivation_of_field`, `not_formallySmooth_of_field`, `no_derivation_mv`, `not_formallySmooth_mv` (Theorem 2.2); `contre_exemple_general`, `not_injective_to_omega`, `omega_kPrimeOf_bijective` (Proposition 2.3).
- `Folder46.lean`: `ker_trivial_of_pow_smul_eq_bot`, `ker_trivial`, `isCohomologous_of_map`, `injective` (Theorem 3.1); `not_isCoboundary_signCocycle` (the hypothesis $p \nmid |\Gamma|$ is needed); `ker_trivial_restrict`, `ker_trivial_sylow`, `injective_restrict`, `injective_sylow_of_isCoboundary`, `injective_sylow_comm` (Theorem 3.2); `mem_span_fixed`, `exists_sum_smul_eq_one`, `isCohomologous_of_restrict_inertia`, `isCohomologous_of_faithful` (Theorem 3.3); `isCohomologous_of_semilocal`, `isCohomologous_of_restrict_inertia_semilocal` (the semi-local version).
- `Folder153.lean`: `omegaBaseChange_surjective`, `Phi_injective_of_injective`, `mem_range_Phi_iff_of_injective`, `coaddition_existsUnique_of_injective`, `comultiplication_existsUnique_of_injective`, `omegaBaseChange_injective`,

`omegaBaseChange_injective_of_flat` (Theorem 4.1); `polya`, `vandermonde`, `coaddition_binom`.

- `Folder77.lean`: `four_dvd_disc_sub_hafnian_sq`, `disc_even_rank`, `not_dvd_rank_two`, `not_dvd_rank_four`, `disc_rank_four` (Theorem 5.1); `two_dvd_det_polar` (odd rank).
- `Folder158.lean`: `etape1` (Lemma 6.1); `proposition1`, `contre_noyau_non_ponctuel` (Theorem 6.2 and the example); `cofiltrant_of_simplyConnected`, `simplyConnected_of_cofiltrant` (page 39 and its converse); `common_ancestor`, `square_of_H`, `isCofiltered_of_H`, `discrete_counterexample` (Theorem 6.3).

The formal proof of Theorem 5.1 follows the Pfaffian route described after it, not the counting argument. The following are not proved here: the geometric statement of folder 2, page 13, on an elliptic curve; the presentation of CK by a single relation; the semi-local Hilbert 90 of folder 46 with wild inertia or a finite residue field; the Sylow injectivity for non-commutative A under the untwisted hypothesis alone; the co-operations of folder 153 under the hypothesis $K_0 \otimes K_0 = K_0$ alone; and the question of folder 158 for infinite categories.

Statement on the use of artificial intelligence

The transcriptions of the folders, the modernised readings, the proofs, the Lean formalisation and a first draft of this text were produced with language models (Claude Opus 5 and Opus 5.5, Anthropic), under the author's direction. The author checked the statements, the proofs and the references, and takes full responsibility for the content. Lean checks the formal proofs mechanically.

Acknowledgements

The facsimiles are published by the University of Montpellier, grothendieck.umontpellier.fr.

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